MCAST

ASSESSMENT AND INTERNAL VERIFICATION FRONT SHEET (Individual Criteria)

(Note: This version is to be used for an assignment brief issued to students via Classter)

Course Title	Bachelor of Science (Honours) in Applied Data Sciences		**Lecturer Name & Surname**	**Frankie Inguanez**	
Unit Number & Title	ITADS-506-2302-Probability and Statistics for Data Science				
Assignment Number, Title / Type	3, Project / Home				
Date Set	04/05/2026	**Deadline Date**	29/05/2026		
Student Name		**ID Number**		**Class / Group**	

Assessment Criteria	Maximum Mark
R&U4 Identify the key features and functionalities of programming libraries used in data science.	5
A&A2 Apply programming concepts and libraries to pre-process, manipulate, and visualize datasets.	7
A&A3 Use libraries to perform statistical analyses, and conducting hypothesis tests.	7
R&U6 Review the choice of statistical models in solving a given data science problem.	5
R&U7 Explain the results of statistical analyses in the context of the underlying data and research question.	5
E&C2 Evaluate the appropriateness and effectiveness of different data visualizations for presenting statistical findings.	10
E&C3 Synthesize the results of statistical analyses while presenting clear, concise, and accurate conclusions using appropriate data visualizations and narratives.	10
Total Mark	49

Notes to Students:

- This assignment brief has been approved and released by the Internal Verifier through Classter.
- Assessment marks and feedback by the lecturer will be available online via Classter (Http://mcast.classter.com) following release by the Internal Verifier
- Students submitting their assignment on Moodle/Turnitin will be requested to confirm online the following statements:

Student's declaration prior to handing-in of assignment

- ❖ I certify that the work submitted for this assignment is my own and that I have read and understood the respective Plagiarism Policy

Student's declaration on assessment special arrangements

- ❖ I certify that adequate support was given to me during the assignment through the Institute and/or the Inclusive Education Unit.
- ❖ I declare that I refused the special support offered by the Institute.

ITADS-506-2301 | Programming for Data Science

ITADS-506-2302 | Probability and Statistics for Data Science

In this assignment you are to investigate the quality of wine using the dataset on the UCI Irvine Repository at https://archive.ics.uci.edu/dataset/186/wine+quality and undertake several experiments to form an opinion. Being a home assignment, it is understood that intelligence tools are used, this is encouraged provided that an audit trail is maintained and that you engage in reflection and personal growth. The focus of this assignment shifts from the provision of a solution script to the evaluation of intelligence tools output.

1. Create a *GitHub* repository making sure you set it as *private*.
2. Add a *Readme.MD* and a *.gitignore* file for *python*. Do not set a *license*.
3. Add your respective lecturer(s) as a *collaborator*. This will allow the evaluation of your gradual work progression. Make sure to follow the guidelines of your lecturer(s) in regard to which email to use for both your repository, and when adding lecturers as collaborators.
4. Structure the repository in the following manner:
    - **doc:** Place the assignment document and any other reference material of importance here.
    - **prompts:** All use of LLMs should be well documented. For each conversation you have with an LLM you are to create a markdown file using the format <Year_month_day_id_purpose.md> such that if you had two conversations on the 21st April 2026 you would have 2026_04_21_01_linear_algebra.md and 2026_04_21_02_correlation.md. Each file should contain a URL to your conversation followed by a reflexive journal entry by yourself on what you understood by the output, how you verified the output (push-back, chain of thought) and how you adapted it.
    - **src:** A single Jupyter notebook that contains the code solution to this assignment. You do not need to download and include the dataset, it is hosted on a server, and you need to research on how to access it remotely. The notebook should be structured to follow a pipeline as demonstrated in class and should contain documentation with observations and interpretations of various test.
5. The root folder of your repository is to include several files:
    - **.gitignore:** This file should be updated to include any IDE folders such as .vscode, any virtual environments such as .venv, any LLM related files such as .claude, whilst also including standard python files and folders which are auto-generated when creating this repository.
    - **requirements.txt:** Upon completion of the assignment, you need to provide a list of python packages and respective versions used in your virtual environment.

This is important for reproducibility. Research how to generate this using pip freeze. Make sure that a separate prompt entry is provided for this task.

- **README.MD:** Update this file to explain the repository structure as specified here. Explain the requirements.txt file.

You are to undertake an experiment that addresses the research question "How to predict the quality of wine?". To address this question:

- Create 1 Jupyter notebook saved under the src folder.
- Create a requirements.txt file in the root folder documenting the packages and versions needed to run the notebook.
- The notebook should be structured following the machine learning pipeline:
    - **Case Study – Wine Quality:** A level 1 markdown with this title. Followed by your full name, group, choice of LLMs. A code cell that addresses the following.
        - **Required Packages**: List down all packages used, and a simple statement what they are needed for.
    - **Problem Definition:** A level 2 markdown section that explains the problem and documents the research question.
    - **Data acquisition:** A level 2 markdown section that documents the dataset as at the UCI Irvine Repository, followed by a series of code cells that download the dataset via URL in a Pandas data frame, previews the structure and rows. Save the combined dataset as JSON file to *wine.json*.
        - **Data Reusability:** Create a copy of the dataset, add a new column with label low if quality is 5 or less, label medium if score is 6 else high. Save this dataset copy to a new CSV file named *modified_wine_quality.csv*. Then reload the file from csv and show the first few records to confirm.
    - **Data exploration:** A level 2 markdown section followed by a series of code cells that review the statistical properties, number of missing values, number of duplicates of the dataset as well as a series of histograms and boxplots. Undertake a correlation analysis for all variables against the quality of wine.
    - **Hypothesis testing:** A level 2 markdown section followed by a series of code cells. Research what a t-test is, then undertake a series of t-tests do determine if there is any statistical difference between red and white wine for the different variables. For each test set the null and alternate hypothesis. Undertake a hypothesis test at 95% confidence level.
    - **Data splitting:** A level 2 mark down section followed by a code cell. In this code cell you are to create an X and y variable. The y variable will be limited to the target variable (quality) representing the quality of wine. The X variable shall have all other variables. These two variables need to be split

in A code cell that splits the dataset using the hold-out strategy, 70% for training and 30% for testing.

- **Regression:** A level 2 markdown section followed by a series of code cells. Here you use the training dataset to create different linear regression models in search for the best model. Based on your evaluation propose the ideal model with justification.
  - **Matrix Operations**: A colleague informed you that they can calculate quality using only features alcohol, volatile acidity, and sulphates. Each of these features has weights 0.6, -1.2, and 0.8. Use matrix operations to find the quality and prepare a data frame storing the actual score and the predicted score. Then display RMSE and R^2 scores.
  - **Linear Algebra:** How can linear algebra be used to calculate a wine quality score? Research ways of calculating wine quality score using linear algebra, this must be done by first discovering the actual weights, then using them to predict quality. Make sure to use the same features alcohol, volatile acidity, and sulphates. Then display RMSE and R^2 scores.
- **Evaluation:** A level 2 markdown section followed by a code cell, using the created model on the test dataset to predict and evaluate the performance of the model.
  - **Linear Algebra:** When you compare linear to regression models, what do you notice about the RMSE and R^2? What can be done to improve them.
- **Observation & Interpretation:** A level 2 markdown section where you document all your findings with a statistical basis and make a recommendation.

VIVA

The assignment is to be worked out in class during the May. A VIVA session shall be held during which random topics covered in this assignment relevant to Programming for Data Science and other random topics relevant to Probability & Statistics for Data Science shall be chosen. The rubric to be applied during the VIVA is provided here below.

Not attempted (0%)	Low (<= 30%)	Medium (>30% <=60%)	High (>60% <=100%)
Did not attend or answer	Superficial evidence given and heavy reliance on LLM	Adequate evidence submitted, some confusion and reliance on LLM	Good, fluent and independent response

Grading Criteria

Programming for Data Science Rubric

RU6 Choose adequate data science packages for data analysis tasks	
Correct justification of each package and why it's needed.	5

AA3 Use data science packages to manipulate a given dataset	
Reading datasets from online server	3
Combining dataframes	2
Saved file as JSON	2

AA4 Modify data through the use of files	
Created a copy of the file	1
Add quality category label	3
Saved file	2
Loaded file and previewed	1

EC2 Develop a program that makes use of advanced data manipulation techniques using data science packages	
Creating dataset for t-test	5
Preparing data for regression	5

RU7 Illustrate how linear algebra concepts are used to perform matrix operations	
Created matrix with required data and produced result	4
Displayed statistics on result	1

AA5 Determine a viable approach for a given data problem

Managed to use linear algebra to extract weights and use them in prediction	4
Displayed statistics on result	1
Valid limitations of linear algebra mentioned in markdown.	2

EC3 Solve a given data science problem through linear algebra	
Managed to explain own work during VIVA	
Did not attend or answer VIVA question	0
Superficial answer given and heavy reliance on LLM demonstrated	3
Adequate answer given during VIVA, some confusion and reliance on LLM	6
Good, fluent and independent response during VIVA	10

Probability & Statistics for Data Science Rubric

Criteria	**Task**	**Marks**
R&U4	Correct setup of repository, Jupyter notebook, and libraries imported	5
A&A2	Correct implementation of pipeline	7
R&U6	Correct explanation of statistical model during VIVA	5
R&U7	Correctly explanation of statistical test during VIVA	5
A&A3	Correct evidence-based implementation of hypothesis tests and statistical analysis during VIVA	7
E&C2	Correct evaluation of statistical models and test during VIVA	10
E&C3	Correct synthesis of experiment and results during VIVA	10